

# Navigators System Architecture

End-to-End Navigation Engine, GNSS Trust Management & Sensor Fusion Pipeline

## Navigators System Architecture

System Version: 1.0.0  
Architecture Pattern: Edge-First Multi-Modal Inertial Navigation

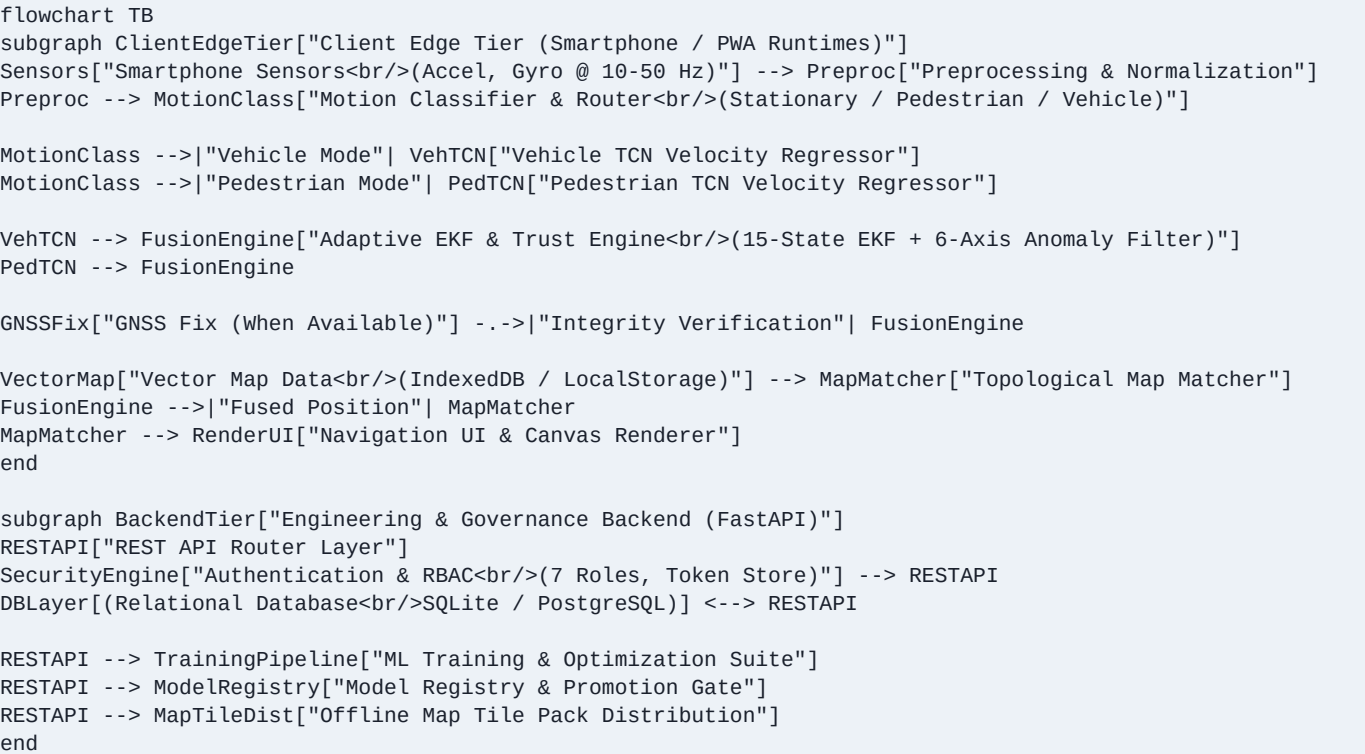
### System Overview

Navigators is an edge-first navigation system engineered to maintain high-accuracy vehicle and pedestrian positioning during Global Navigation Satellite System (GNSS) degradation or complete outage.

The architecture separates execution into two principal tiers:

- **Client-Side Edge Runtime:** Autonomous positioning, motion classification, deep velocity inference, Kalman filtering, and map matching executing inside client runtimes (web browser, PWA, or smartphone process).
- **Engineering & Control Backend:** A FastAPI service for dataset ingestion, training pipeline orchestration, model candidate registration, vector map pack serving, and security governance.

### Architectural Block Diagram



### Tier Responsibilities and Boundaries

#### Edge Execution Tier

- **Zero Internet Dependence:** Navigation state propagation, motion inference, EKF filtering, and map matching execute strictly locally on client hardware.
- **Sensor Ingestion:** Ingests tri-axial acceleration and angular velocity at frequencies up to 50 Hz.

- **State Estimation:** Manages the 15-state error formulation tracking 3D position, 3D velocity, 3D attitude, accelerometer biases, and gyroscope biases.

### **Backend Control Tier**

- **Dataset Registry:** Ingests, validates, and manages immutable datasets with full provenance.
- **Model Registry:** Manages model staging, validation metrics verification, and team-admin approval workflows before promoting model candidates to production status.
- **RBAC & Governance:** Enforces 7 distinct system roles and atomic permission policies across dataset and model deployment operations.